

Improving Cultural Knowledge Coverage in LLMs via Region-Aware RAG

An Extension of BLEnD — Mid-term Report: Data, Knowledge Base and Baseline

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Abstract

We extend the BLEnD benchmark (Myung et al., NeurIPS 2024) with inference-time, region-aware retrieval to test whether retrieval narrows the gap between high- and low-resource cultures. This mid-term report covers the first half of the project: (i) the evaluation subset (5 countries, 9 country–language settings, 4,500 short-answer questions), (ii) an independently built, contamination-safe English cultural knowledge base (720 evidence cards), and (iii) a full replication of BLEnD’s zero-shot baseline protocol on three free, locally run open-weight LLMs (27,000 model answers, scored with BLEnD’s official scorer). The baseline reproduces the paper’s central finding: accuracy falls from 75% (US) to 8% (Ethiopia, in Amharic) on identical questions, and low-resource cultures are answered better in English than in their own language.

1 Scope

BLEnD contains 500 everyday-culture question templates (food, sports, family, education, holidays/celebrations/leisure, work-life). Each is asked about 16 countries, answered by five native annotators per country. We use the **Short-Answer Question (SAQ)** track for **5 countries** that span the resource spectrum. Every non-English country is evaluated twice: with the question in its local language and in English. This gives **9 country–language settings**.

Table 1: Countries, languages and question counts. *Scorable* = questions kept by BLEnD’s official filter (dropped when ≥ 3 annotators said “no answer / not applicable” or ≥ 5 said “I don’t know”). Resource class of the language follows Joshi et al. as used in the BLEnD paper.

Country	Local language	Resource	Settings run	Questions / setting	Scorable
United States	English	high	en	500	461
Spain	Spanish	high	es, en	500	469
South Korea	Korean	mid-high	ko, en	500	475
Azerbaijan	Azerbaijani	low	az, en	500	469
Ethiopia	Amharic	low	am, en	500	472
Total: 5 countries, 6 languages			9 settings	4,500	4,231

The SOP also lists Assam. It was dropped from this phase because no Assam knowledge base was built. Table 2 breaks the scorable questions down by category. Every setting of a country shares the same scorable set, so the English and local-language versions are directly comparable.

2 Cultural Knowledge Base (completed)

For the RAG phase we built one knowledge base (KB) per country. It consists of short English *evidence cards*, each giving one documented cultural fact with its source. To avoid the contamination problem reported in SemEval-2026 Task 7, **no BLEnD material was used**. Sources are

Table 2: Scorable questions per category (out of the 500 templates: Food 105, Sport 88, Holidays 92, Education 84, Work-life 68, Family 63).

Country	Food	Sport	Holidays	Education	Work-life	Family	Total
US	100	76	81	81	65	58	461
Spain	102	82	78	81	65	61	469
South Korea	102	86	80	82	64	61	475
Azerbaijan	103	82	75	83	67	59	469
Ethiopia	101	77	86	83	67	58	472

institutional: UNESCO intangible-heritage pages, national statistics offices, education and labour ministries, national tourism boards, the US Library of Congress and similar. Every card records its category, topic, region/sub-region, time scope, source URL and a source-quality label.

Table 3: Knowledge-base cards per country and category (target: 25 per category).

Country	Food	Sports	Family	Education	Holidays/Leisure	Work-life	Total
US	25	25	25	25		25	150
Spain	25	25	25	25		25	150
South Korea	21	19	25	19		25	130
Azerbaijan	25	25	25	25		25	150
Ethiopia	25	22	25	23		25	140
Total							720

Retrieval will index the English text of each card (`text_en`). For a local-language question, an English rendering is used *only* as the search query; the LLM still receives the original-language question.

3 Baseline: Experimental Setup

Protocol (replicated from BLEND). Every question is asked with the paper’s two prompts, written in the same language as the question:

- **inst-4** (direct): “Read the following question and provide a single answer without any explanations. Question: {*q*} Answer:”
- **pers-3** (persona): “You are a person from {*country*} who is trying to explain your country’s culture to a foreigner. Answer the following question, providing a single answer without any explanations. {*q*}”

Decoding is greedy (temperature 0), with at most 512 new tokens, as in the paper. No retrieval and no fine-tuning are used. Each model produced $9\text{settings} \times 500 \times 2\text{prompts} = 9,000$ answers, **27,000 in total**.

Models. All three are free, open-weight and run locally. No paid API was used.

Model	Parameters	Format	Licence
Gemma 3 4B-it (<code>gemma3:4b</code>)	4.3B	4-bit (Q4_K_M)	Gemma terms
Mistral-7B-Instruct v0.3 (<code>mistral</code>)	7.2B	4-bit (Q4_0)	Apache-2.0
Qwen2.5-3B-Instruct (<code>qwen2.5:3b</code>)	3.1B	4-bit (Q4_K_M)	Qwen research

Inference ran with Ollama on an institute workstation (NVIDIA RTX 3060, 12 GB). **Deviation from the SOP:** the SOP planned larger models (Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, Llama 3.1) on a 32 GB GPU. The available machine has 12 GB of GPU memory, under 5 GB of

free disk, and a firewall that blocks model downloads. We therefore evaluate 3–7B 4-bit models, a *resource-constrained* setting where retrieval has the most room to help.

Scoring (official, unchanged). We use BLEnD’s own `soft_exact_match` (`evaluation/exact_match.py`). An answer is correct if any annotator answer appears in it after lemmatisation/stemming. The tools are spaCy (English), Okt/konlpy (Korean), the aznlp stemmer (Azerbaijani) and Spark NLP lemmatisers (Spanish, Amharic). English answer variants are accepted for local-language questions. The released inference and evaluation wrappers do not run as released, so we call the scoring function directly. We report **SEM-B** (binary accuracy, %) averaged over the two prompts, exactly as the paper does.

4 Baseline Results

Table 4: Baseline SAQ accuracy (SEM-B, %), averaged over `inst-4` and `pers-3`. Right block: published scores of two models from the BLEnD paper (Tables 10/11), for reference.

Country	Lang.	Ours (4-bit, 3–7B)				BLEnD paper	
		Gemma3-4B	Mistral-7B	Qwen2.5-3B	Mean	Aya-101	Llama-3.1-70B
US	en	72.1	84.9	67.8	74.9	53.4	84.9
Spain	es	58.8	61.3	48.6	56.2	45.8	75.4
	en	41.2	65.6	45.7	50.8	35.7	59.4
South Korea	ko	54.3	47.9	41.9	48.0	32.8	65.3
	en	43.7	60.6	38.7	47.7	30.7	62.1
Azerbaijan	az	29.4	27.7	13.6	23.6	35.8	59.5
	en	45.7	59.8	37.1	47.5	31.9	59.5
Ethiopia	am	14.4	3.7	5.6	7.9	17.8	17.6
	en	30.6	46.3	25.5	34.1	26.4	44.6
Mean, local language		45.8	45.1	35.5	42.1		
Mean, English		46.7	63.4	43.0	51.0		

Table 5: Accuracy (%) by category, pooled over all 9 settings and both prompts.

Model	Food	Sport	Holidays	Education	Work-life	Family
Gemma3-4B	38.8	43.8	36.2	49.1	52.9	41.3
Mistral-7B	46.8	52.5	46.1	55.8	57.1	47.5
Qwen2.5-3B	30.7	33.4	30.9	43.8	44.2	35.5

4.1 Observations

- The cultural gap is large and reproduces the paper.** In the local language, the 3-model mean falls from 74.9% (US) to 56.2% (Spain), 48.0% (South Korea), 23.6% (Azerbaijan) and 7.9% (Ethiopia). The paper’s 16-model averages show the same ordering (US 79.2, Spain 69.1, Ethiopia 12.2). High-resource local settings average 53–65%; low-resource ones average 10–22%.
- Low-resource cultures are answered better in English.** Azerbaijan improves from 23.6% to 47.5% and Ethiopia from 7.9% to 34.1% when the same question is asked in English. For Spanish and Korean the local language is as good or better (e.g. Gemma: 54.3% Korean vs 43.7% English). This matches BLEnD’s finding exactly.

3. **Amharic is nearly unsolved by small models** (3.7–14.4%). Many outputs are not fluent Amharic, e.g. repetitions of the prompt.
4. **Our models sit between the paper’s Aya-101 and Llama-3.1-70B.** Mistral-7B matches Llama-3.1-70B on US and English settings. All our models remain well below it on non-English local-language settings.
5. **Category effects match the paper:** food and holidays are hardest, while education and work-life are easiest, for all three models (Table 5).

4.2 Caveats of the official metric

- **Verbosity.** Mistral ignores “single answer” more often: median answer length is 13 words, against 2 for Gemma and Qwen. Since the metric checks whether a gold answer is *contained* in the response, longer answers have more chances to match. Part of Mistral’s lead may come from this.
- **Language switching under the persona prompt.** With **pers-3**, Gemma often answers English questions about Spain in Spanish (“Naranja” for “orange”). English settings accept only English gold answers, so its Spain-English score drops from 48.4% (**inst-4**) to 33.9% (**pers-3**).
- **Morphologically rich languages.** Lemma/stem matching under-counts correct answers in morphologically rich languages, as noted in the SOP.
- **Software versions.** Two scorer libraries are newer than the paper’s because of Python/Java compatibility: Spark NLP 5.5.3 (paper: 5.3.3) and pandas 2.2.3 (2.2.2). The lemmatiser models and the scoring code are unchanged.

5 Next Steps

1. **Always-RAG:** hybrid BM25 + dense (bge-m3) retrieval over the country’s KB, top-*k* cards prepended to the same prompts. Then run the same 27,000-answer protocol and compute the per-setting change against Table 4.
2. **Adaptive-RAG:** a training-free gate (retrieval score / category prior) choosing per question between the RAG and baseline answers. It needs no extra model calls.
3. **Cross-lingual retrieval:** machine-translated English queries for Azerbaijani and Amharic questions, compared against the oracle English query.
4. **Category-aware analysis** of where retrieval helps or hurts, from the per-question logs.

Reproducibility. The code (`rag_pipeline/`), all prompts sent, all 27,000 responses and per-question scores are in the project repository (https://github.com/SidRai-247/Blend_ML_Project).